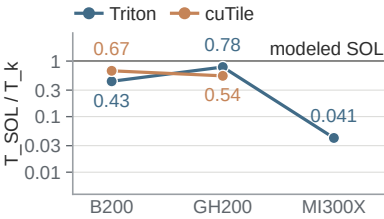


A Matrix operand delivery (matmul, FP32)



T_SOL: TF32-class MMA term (MI300X: XF32); compute-bound

NVIDIA NCU, profiled run()

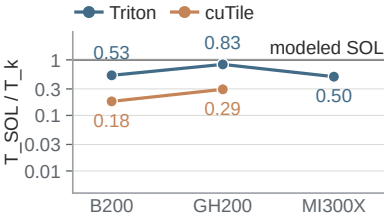
	B200		GH200	
	Triton	cuTile	Triton	cuTile
MMA instruction family	tcgen05	tcgen05	WGMMA	WGMMA
TMA load bytes (GB)	21.5	10.7	21.5	21.5
STS / LDSM instr. (M)	0.13 / 0.00	0.13 / 0.00	0.26 / 0.00	21.23 / 20.97

MI300X Triton (static ISA): TensorDescriptor loads lowered to 96 scalar 32-bit global_load_dword; MFMA utilization 4.5% (rocprof-compute)

Interpretation: on GH200, cuTile adds one STS and one LDSM per WGMMA, consistent with an on-chip operand re-layout; B200 feeds tcgen05 via TMA.

Confounders: Triton reads B pre-transposed (outside timing); cuTile tiles 256×256 (B200) vs. 128×128 (GH200); cuda-tile 1.3.0 vs. 1.5.0.

B Indexing overhead (destindex, INT8)



T_SOL: memory-only (Q / BW)

NVIDIA NCU, profiled run()

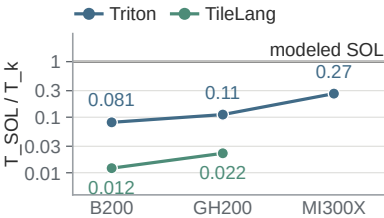
	B200		GH200	
	Triton	cuTile	Triton	cuTile
Executed instructions (M)	5.2	90.3	5.2	92.9
Global store instr. (M)	0.13	2.05	0.13	2.05
Store width	128-bit	8-bit	128-bit	8-bit

MI300X Triton (static ISA): each row is stored with 4 per-lane buffer_store_byte, no vector store

Interpretation: for the same 2.05 M store sectors, cuTile issues 16× more (byte-wide) stores and about 17× more instructions than Triton on both NVIDIA GPUs.

Confounders: NVIDIA counts are dynamic, MI300X counts static; the memory-only target does not model index arithmetic.

C Memory access and latency hiding (1d_conv, FP16)



T_SOL: max(FP16-MMA, HBM) terms; compute-bound on B200 and MI300X, memory-bound on GH200

NVIDIA NCU, profiled run()

	B200		GH200	
	Triton	TileLang	Triton	TileLang
L1 global-load sectors (M)	147	1091	210	2181
Occupancy ach. / limit (%)	24.6 / 25.0	6.2 / 18.8	24.8 / 25.0	18.7 / 18.8
Issue active (%)	57.8	3.9	61.5	22.4

MI300X Triton (static ISA, rocprof): buffer_load_ushort loads; 66% of VALU instr. are INT32. PyTorch (MIOpen implicit GEMM + 3 transposes) is not part of T_SOL / T_k.

Interpretation: with the same 16-bit loads and 0.67 GB of DRAM reads, TileLang touches 7–10× more L1 sectors; on B200 it reaches 6.2% of an 18.8% occupancy limit.

Confounders: TileLang uses a different kernel body on Hopper; no controlled experiment isolates the MI300X difference.